DBMS REVIEW 1

Design and Implementation of a Database Management System for Scholarship and Financial Aid Management System

Problem Identification • Literature Survey • Project Planning & SRS

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What We'll Cover Today

01

Introduction & Motivation

Why a dedicated scholarship & aid database matters

02

Problem Statement

The core issue this project addresses

03

Objectives & Scope

What the system will and won't do

04

Literature Survey

Existing systems, approaches & research gap

05

Project Planning

Timeline, methodology & milestones

06

Software Requirements Specification

Functional & non-functional requirements

Why This Project?

Scholarships and financial aid are lifelines for thousands of students, but the processes that manage them are often **fragmented, manual, and error-prone.**

Institutions today rely on spreadsheets, paper forms, and disconnected portals to track eligibility, applications, disbursement, and renewals — creating delays, data loss, and lack of transparency for both students and administrators.

A well-designed DBMS brings a single source of truth: consistent data, faster processing, auditability, and fair, rule-based allocation of financial aid.

3+

Disconnected systems typically used per institution

High

Risk of duplicate or inconsistent aid records

Manual

Eligibility checks & renewal tracking today

PROBLEM IDENTIFICATION

Problem Statement

Existing scholarship and financial-aid workflows lack a centralized, rule-driven database, resulting in inconsistent eligibility evaluation, delayed disbursement, duplicate applications, and limited visibility for students, administrators, and auditors.

Fragmented Data

Student, financial, and academic records live in separate, unlinked files or systems.

Manual Eligibility Checks

Prone to human error and inconsistent application of scholarship criteria.

Poor Traceability

No reliable audit trail for who approved what aid, and when.

Slow Disbursement

Application-to-payout cycle is delayed by manual verification steps.

Objectives & Scope

Project Objectives

- Design a normalized relational database for students, scholarships, donors, and disbursements
- Automate eligibility evaluation using rule-based criteria stored in the database
- Enable transparent tracking of application status and fund disbursement
- Provide role-based access for students, reviewers, and administrators
- Generate audit-ready reports for compliance and decision-making

Scope of Review 1

- ✓ Problem identification & motivation
- ✓ Literature survey of existing systems
- ✓ Requirement gathering approach
- ✓ Preliminary project plan (SRS-ready)

Out of scope for this review: full implementation, UI development, and final testing — covered in Review 2 & 3.

Review of Existing Systems

System / Study	Approach	Strength	Limitation
College spreadsheet-based tracking	Excel/Google Sheets maintained manually	Low cost, easy to start	No validation, high error rate, no concurrency control
Generic Student Information Systems (SIS)	Centralized academic records management	Robust academic data model, widely adopted	No dedicated scholarship / aid eligibility engine
Govt. scholarship portals (e.g. National Scholarship Portal-style systems)	Web application with central database backend	Standardized eligibility rules, wide reach	Rigid to institution-specific criteria, limited customization
Research on rule-based eligibility engines in DBMS	Stored procedures / triggers enforce eligibility rules	Consistent, automatable decision logic	Needs careful schema design; complexity grows with rule variety

Research Gap & Opportunity

1

No Unified Schema

Student, academic, financial, and donor data are rarely modeled together relationally.

2

Static Eligibility Logic

Criteria are hardcoded or manually applied instead of rule-driven and configurable.

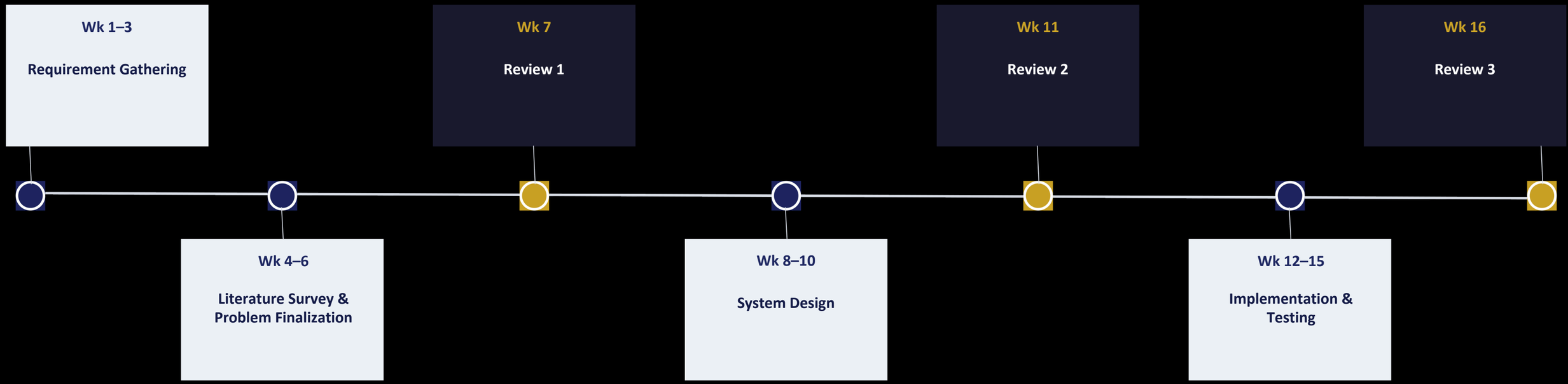
3

Weak Audit Trail

Few systems log the full lifecycle of an application from submission to disbursement.

Opportunity: Build a purpose-fit DBMS that unifies these three gaps into one coherent, rule-driven, auditable platform for scholarship and financial aid management.

Methodology & Timeline



Review 1 Deliverables Checklist

✓	**Problem Identification Report**	Documented problem statement with motivation and impact
✓	**Literature Survey**	Minimum 5 sources reviewed with comparison table
✓	**Draft SRS Document**	Functional & non-functional requirements outlined
✓	**Preliminary ER Model**	High-level entities identified (Student, Scholarship, Application, Disbursement)
✓	**Project Plan / Timeline**	Phase-wise plan mapped to Review 2 and Review 3

Functional Requirements

1 Student & User Management

- Register/authenticate students, reviewers, admins
- Role-based access control (Student / Reviewer / Admin)

2 Scholarship Catalog

- Define scholarship schemes with eligibility rules
- Support merit-based, need-based & donor-funded categories

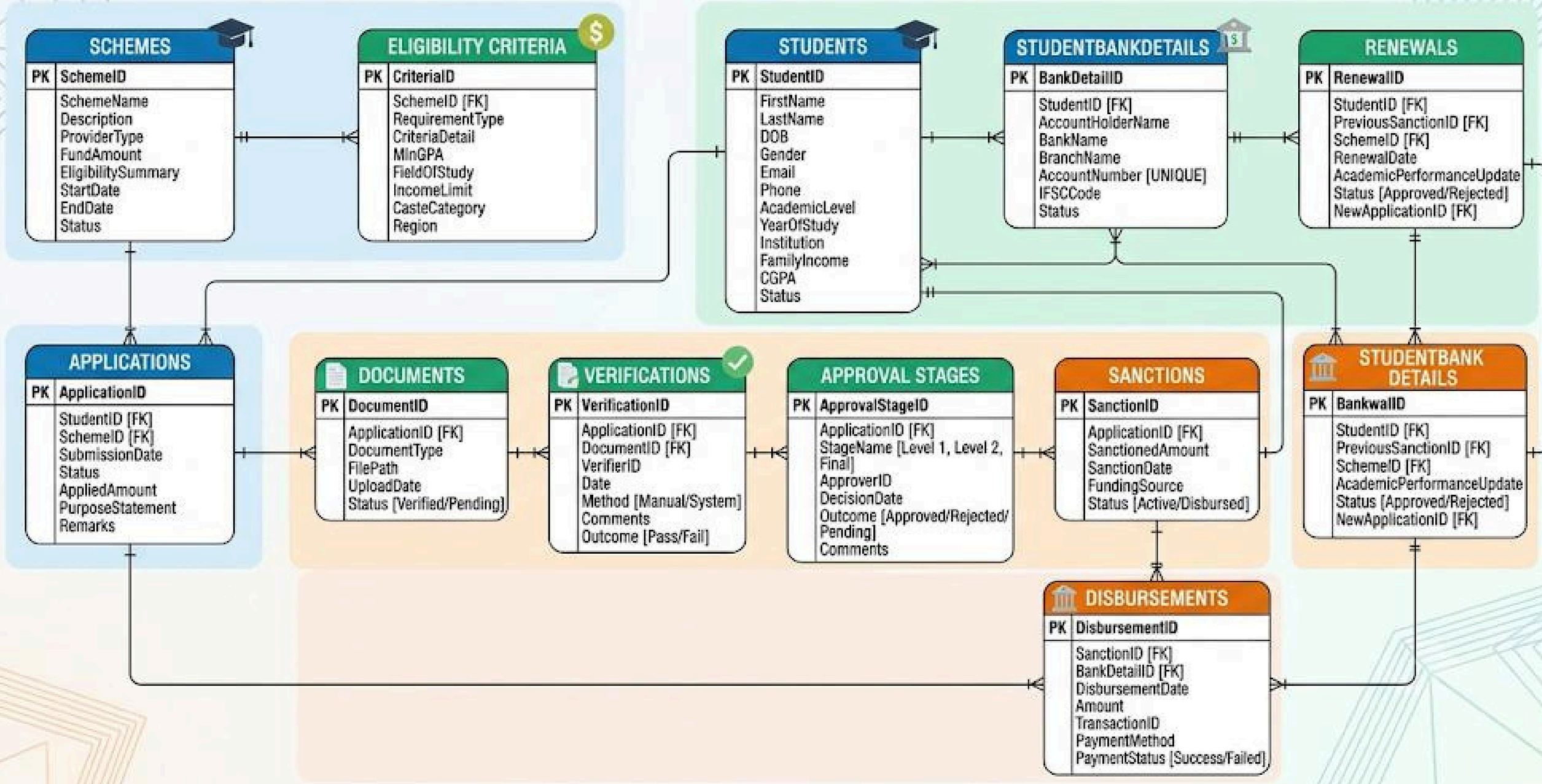
3 Application Processing

- Submit, track, and update application status
- Automated eligibility validation against stored rules

4 Disbursement & Reporting

- Record and track fund disbursement per student
- Generate audit and compliance reports

SCHOLARSHIP AND FINANCIAL AID MANAGEMENT SYSTEM



NonFunctional Requirment-Alingnment

Security

Encrypted credentials, role-based access, data privacy compliance

Reliability

Consistent, ACID-compliant transactions for aid disbursement

Scalability

Handle growing student & scholarship records across semesters

Usability

Simple workflows for students and administrators alike

Performance

Fast query response for eligibility checks & report generation

Auditability

Full traceable history of every application and disbursement

Preliminary Data Entities

High-levelentitiesidentifiedfortheERmodeltobedetailedinReview2.

Student	Scholarship	Application	Disbursement
Student_ID (PK)	Scholarship_ID (PK)	Application_ID (PK)	Disbursement_ID (PK)
Name, DOB, Program	Name, Type, Amount	Student_ID, Scholarship_ID (FK)	Application_ID (FK)
Income Category, CGPA	Eligibility_Rules	Status, Submit_Date	Amount, Date, Mode

Recap & Next Steps

- Identified a clear, real-world problem in scholarship & aid management
- Surveyed existing systems and located a concrete research gap
- Outlined a phased project plan through Week 16
- Drafted preliminary SRS: functional, non-functional & data entities

Next: Toward Review 2

- → Finalize ER diagram & relational schema
- → Normalize tables up to 3NF
- → Begin implementation & prototype demo

Thank You